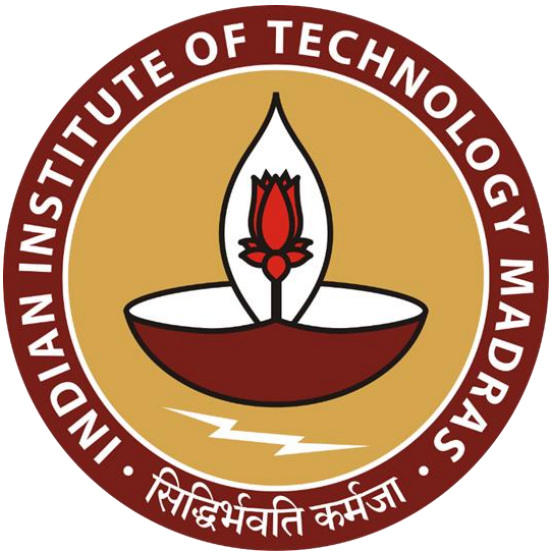
Optimizing Saree Retail Operations with Sales and Inventory Analytics at S.R Trading Co.: A Data-Driven Approach

Mid Term Report

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Contents

1. **Executive Summary:** 1
2. **Proof of originality of Data:** 1

Official Documents: 2

Data: 2

3. **Metadata and Descriptive Statistics:** 2

- **Metadata**: 2
- **Descriptive Statistics:** 3

4. **Detailed Explanation of Analysis Process:** 5
5. **Results and Findings:** 6

1. Executive Summary:

This report represents a pivotal point in S.R. Trading co' effort to improve inventory management. This project offers data-driven solutions to S.R. Trading co ' inventory and profitability problems, in line with the goal of the Business Data Management Capstone project.

Three months of data collecting, from Jan to Mar 2025, allowed for a thorough analysis of the shop's sales and inventory trends. Mr. Anil thakur, assisted me. The owner was hesitant to share the turnover of his business.

The report presents a thorough analysis of the data with the help of MS Excel and Google Sheets, including Pareto analysis, sales analysis. The Pareto analysis highlights the significance of concentrating on these core products by showing how a tiny subset of high-performing items contributes significantly to the store's revenue.

With the help of the data analysis done, the report tackles the problem statements that were discussed with the owner.

The report's conclusion reiterates the possibility of significant operational improvements and offers tactics to improve inventory control, raise overall profitability, and improve sales performance. S.R. Trading co. can seek to optimize its operations, and create a path for long-term, sustainable growth in the sarees sector with the use of data-driven insights.

2. Proof of originality of Data:

After all the discussions with the owner and getting his approval to collect data of his business, the data collection process started. The store doesn't keep any data in a record or tabulated form. Hence, to get data in a proper format, the data was collected in the form of bills was done. The data for each Bill was then converted into an Excel spreadsheet.

Official Documents:

Name	Type	Link
Authentication letter of the business approving the use of data	Image	Authentication letter
Interaction video with the business owner	Video	Interaction Video
Shop Pictures / Video	Image / Video	Shop Picture

Data:

Raw Data	Images	Raw Data
Data Analysis for Mid-Term	Excel Spreadsheet	SpreedSheet

3. Metadata and Descriptive Statistics:

- **Metadata:**

1) Data source: The data was initially collected in the form of Bill copies of invoices. As S.R. Trading co. does not have a proper billing system.

2) Duration of Data Collection: Data collection was done for a period of 3 months, from Jan 2025 to Mar 2025.

3) Number of SKUs : 44 unique saree SKUs across 5 categories

4) Method of Data Collection: The data was collected by WhatsApp Weekly/ Daily and visiting shop.

5) Data Cleaning and Conversion: Data collected in a Bill form was converted to an Excel spreadsheet which was further cleaned.

6) Data analysis tools: The majority of analysis is done with the help of Microsoft Excel.

7) The following table explains the columns in the datasets:

DATE	The date of the transaction
Order No.	Internal order reference number
Invoice No.	Bill/invoice number for the transaction
Party Name	Name of the supplier (purchase) or customer (sales)
Place of Supply	Geographic location of the party
Description of Good	Name/type of the saree SKU
Category	5 Product category (Print, Handwork, Others, Brush, Woven)
HSN/SAC Code	Harmonized System Nomenclature code for GST purposes
Quantity (Pcs.)	Number of pieces in the transaction
Price	Unit selling/purchase price (₹)
Amount	Gross amount before discount
Discount	Discount amount applied
Total Amount	Net transaction value after discount

- **Descriptive Statistics:**

Sales:

Metric	Qty	Price	Total Amount
mean	14.35	938.80	10637.21
Median	7.50	837.50	6922.95
Mode	6.00	425.00	4300.00
Std deviation	18.61	604.31	9402.87
Min	2.00	395.00	1552.00
Max	115.00	3450.00	47725.00
Q1 (25%)	5.00	507.50	3530.00
Q3 (75%)	14.75	950.00	15075.00
Unit Sold	775.00	-	-

- A total of 775 units were sold across 54 transactions over three months, averaging approximately 14.35 Qty per transaction.

- The mean (average) reflects the typical transaction size; an average sale involves about 14 pieces at ₹939 per unit.

- The median is lower than the mean (7.5 pieces, ₹838), indicating that most individual transactions are smaller, while a few large bulk orders pull the mean upward — a right-skewed distribution.

- The standard deviation measures the spread of values around the mean; the high standard deviation in Quantity (18.61) and Total Amount (₹9,403) confirms wide variability in order sizes, with some orders being very large and others quite small.

- The Minimum value (Min) represents the smallest single transaction recorded across the three months i.e. 2 pieces at a unit price of ₹395. The Maximum (Max) denotes the largest single transaction i.e. 115 pieces in one order and the highest recorded price of ₹3,450.

- Unit sold ranges from 2 to 115

- The highest-revenue month was February 2025 with ₹2,58,926, followed by January (₹2,27,945) and March (₹87,538).

- The highest-revenue product was Saree (EBR) with a total of ₹84,150 across 43 units, followed by Chamundi print (Flower) at ₹47,725 (115 units).

- The average unit price was ₹938.80, ranging from a minimum of ₹395 to a maximum of ₹3,450.

Purchases:

Metric	Qty	Price	Total Amount
mean	13.88	974.18	10322.42
Median	7.00	860.00	6300.00
Mode	6.00	900.00	4300.00
Std deviation	19.09	622.83	9313.79
Min	2.00	395.00	1520.00
Max	115.00	3450.00	45338.75
Q1 (25%)	5.00	500.00	3301.25
Q3 (75%)	14.00	975.00	14440.00
Unit Purchased	680.00	-	-

- A total of 680 units were purchased across 49 transactions.

- The mean (average) reflects the typical transaction size; an average sale involves about 13 pieces at ₹974.18 per unit.

- The median is lower than the mean (7 pieces, ₹860), indicating that most individual transactions are smaller, while a few large bulk orders pull the mean upward — a right-skewed distribution.
- The standard deviation measures the spread of values around the mean; the high standard deviation in Quantity (19.09) and Total Amount (₹9,313) confirms wide variability in order sizes, with some orders being very large and others quite small.
- Purchases were sourced from multiple suppliers including Madhu Creations, Radhika Prints, and others in Chandni Chowk.
- The Print category was the most purchased by volume (290 pcs), aligned with it also being the highest-selling category.

Category-wise Sales Summary:

Category	Unit Sold	Revenue (₹)	Average Price (₹)	Revenue %
Print	349	₹ 2,09,475.10	₹ 829.09	36.47%
Others	141	₹ 1,74,900.00	₹ 1,495.00	30.45%
Handwork	237	₹ 1,57,671.60	₹ 844.38	27.45%
Brush	38	₹ 27,062.50	₹ 693.00	4.71%
Woven	10	₹ 5,300.00	₹ 530.00	0.92%

- The top three categories (**Print**, **Others**, and **Handwork**) are contributing **94%** of total revenue.
- Print is the Category which generates the most revenue i.e. **₹ 2,09,475.10**
- Although Prints Generates the most revenue but **Avg Price** of Others categories are the Approx **80%** Higher from Avg of Prints, which shows that it is a Premium Product.

4. Detailed Explanation of Analysis Process:

The analytical process is designed to address the core operational challenges faced by S.R. Trading Co., namely inventory mismanagement, low profitability, reactive purchasing, and absence of data-driven decision making. The goal is to derive actionable insights from three months of manually collected transactional data.

4.1 Data Collection and Preparation

Data collection was conducted through repeated WhatsApp Weekly/ Daily and by visiting shop in Chandni Chowk. Physical bills for all purchases and sales from January to March 2025 were reviewed,

and the relevant data fields were manually transcribed into Excel spreadsheets. Separate sheets were maintained for purchase data and sales data, and each entry was cross-checked against the corresponding bill to ensure accuracy.

4.2 Data Cleaning and Transformation

After entry, the data was cleaned to handle inconsistencies such as missing values, duplicate entries, and formatting irregularities in dates, product names, and numeric fields. Categorical labels for product types were standardized across both the purchase and sales sheets to allow consistent cross-analysis. Numeric fields (Quantity, Price, Amount, Discount, Total Amount) were validated and converted to appropriate data types.

4.3 Category-wise and Product-level Analysis

Products were grouped into five categories — Print, Handwork, Others, Brush, and Woven. For each category, total quantity sold, total revenue, and average price were calculated. Additionally, a product-level breakdown was performed to identify top-selling SKUs by both quantity and revenue, informing procurement priorities.

4.4 Pareto Analysis

A Pareto analysis was conducted on category-level revenue to identify which categories contribute most significantly to overall sales income. The analysis reveals that Print and Others categories together account for approximately 67% of total revenue, providing a clear focus area for inventory investment.

4.5 Sales Trend Analysis

Monthly revenue and volume trends were computed and visualized using bar charts. This analysis identifies seasonal patterns and demand fluctuations across the three-month period, which can inform future stock planning and supplier ordering timelines.

4.6 Purchase vs. Sales Inventory Analysis

A comparison of quantities purchased versus quantities sold by category was performed to identify potential overstock or understock situations. This analysis helps pinpoint areas where procurement needs to be optimized to reduce excess inventory carrying costs or to prevent stockouts.

4.7 Customer Revenue Concentration Analysis

A customer-level revenue breakdown was carried out using pie charts and summary tables. This analysis highlights the degree of dependence on individual B2B clients and helps evaluate the credit exposure risk associated with each customer relationship.

5. Results and Findings:

5.1 Pareto Analysis Revenue by Category:

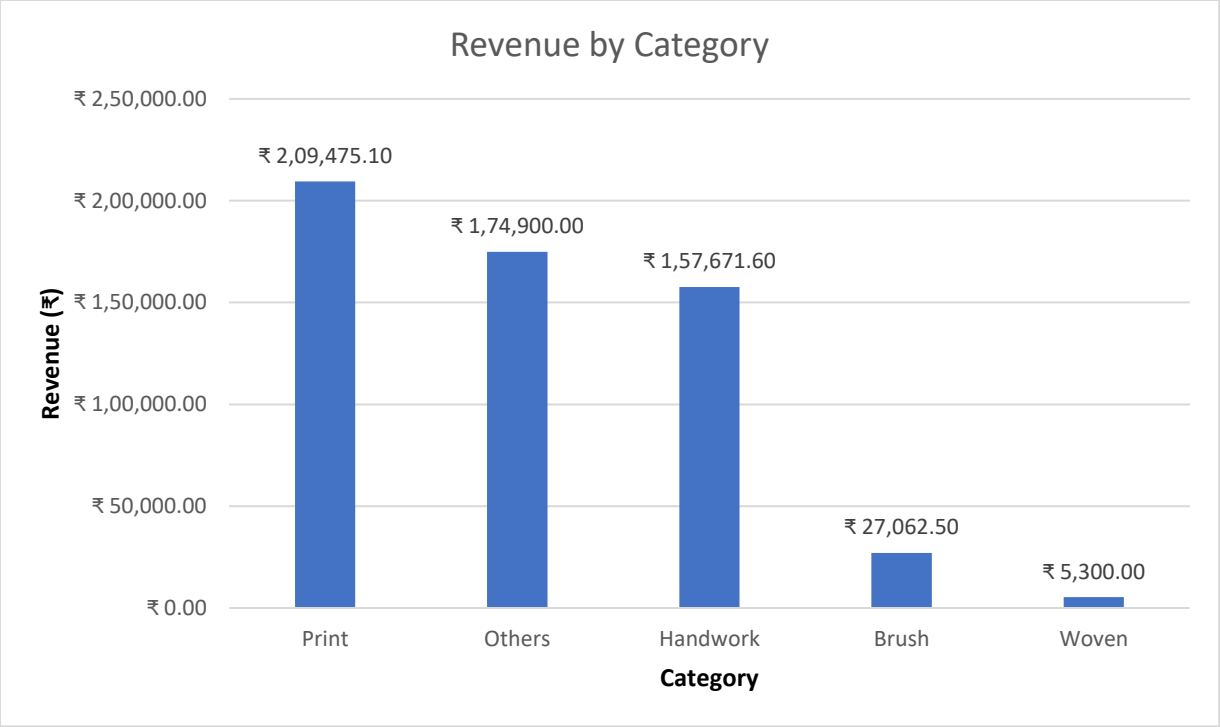


Figure 1: Revenue by Category (Jan–Mar 2025)

The Pareto chart below illustrates revenue contribution by saree category. The Print category is the single largest contributor at ₹2,09,475 (36.5%), followed by Others at ₹1,74,900 (30.5%) and Handwork at ₹1,57,672 (27.4%). Together, these three categories generate over 94% of total revenue, while Brush and Woven represent marginal contributions.

This finding suggests that inventory investment should be heavily concentrated in Print, Others, and Handwork sarees. The Woven category (only 10 units sold, ₹5,300 revenue) may not justify the shelf space or capital it occupies, while Brush items serve a complementary but secondary function

5.2 Monthly Sales Trend

The bar chart below shows the month-wise revenue performance across the three-month data period.

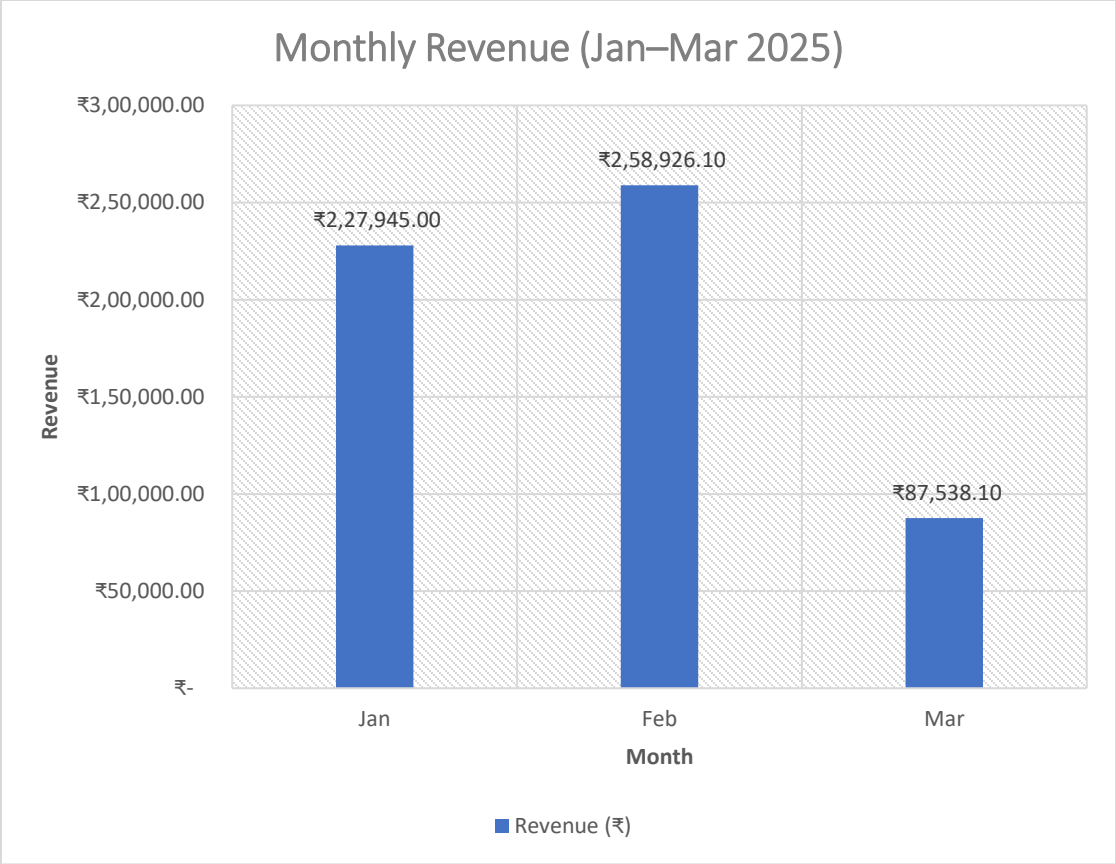


Figure 2: Month-wise Revenue (Jan–Mar 2025)

February 2025 emerges as the strongest month with ₹2,58,926 in revenue and 434 units sold — nearly double January's volume. This uplift may reflect demand from the spring wedding and festival season. March shows a significant drop to ₹87,538 and only 98 units, which may be attributed to the data being captured only partially (up to March 28). January maintained steady performance at ₹2,27,945.

This monthly pattern suggests the business should build up Print and Handwork inventory ahead of January–February to capitalize on the peak demand window.

5.3 Top Products by Revenue

The horizontal bar chart below presents the eight highest-revenue products over the three-month period.

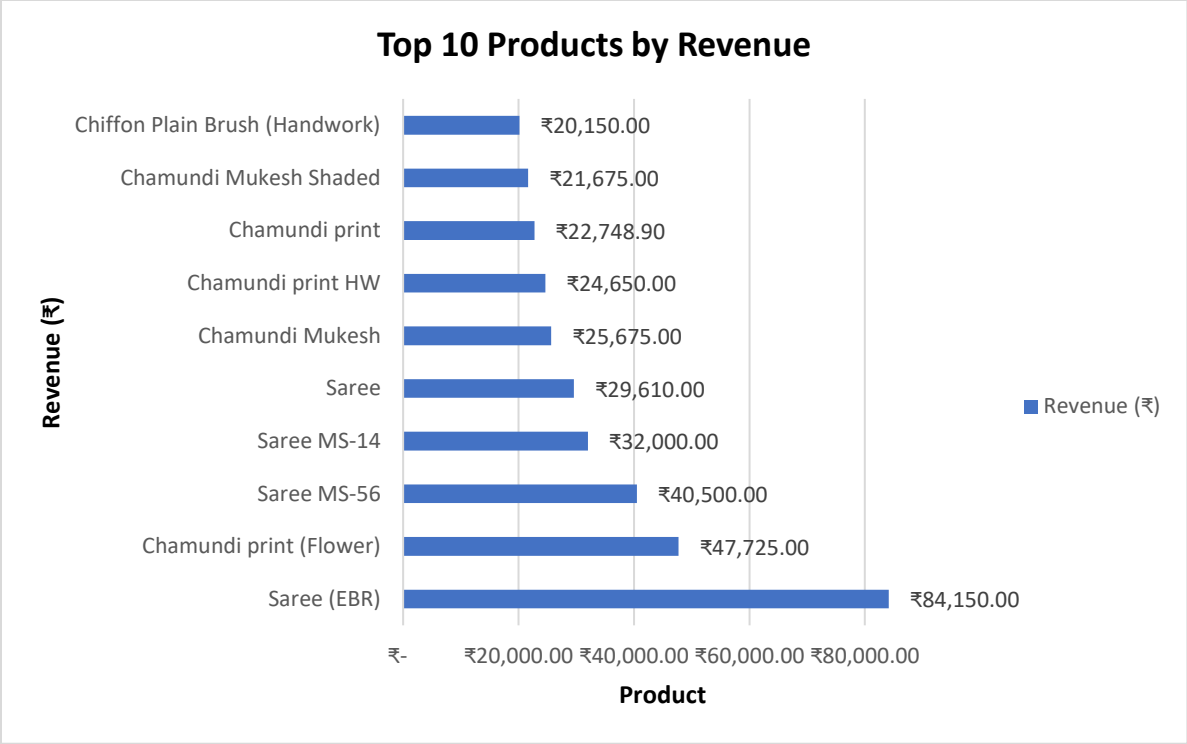


Figure 3: Top 10 Products by Revenue (Jan–Mar 2025)

Saree (EBR) is the clear revenue leader at ₹84,150 despite selling only 43 units, indicating a high unit price point. Chamundi print (Flower) leads in quantity (115 pcs) while generating ₹47,725. Saree MS-56 and Saree MS-14 are also strong performers, contributing ₹40,500 and ₹32,000 respectively. The Chamundi product line (Mukesh, Print, Brush variants) collectively represents a major portion of recurring sales volume.

Ensuring consistent stock availability of these top SKUs — particularly Saree (EBR) and the Chamundi range — should be a priority for inventory management.

5.4 Customer Revenue Distribution

S.R. Trading Co. currently serves four B2B customers. The pie chart below shows the revenue share attributable to each.

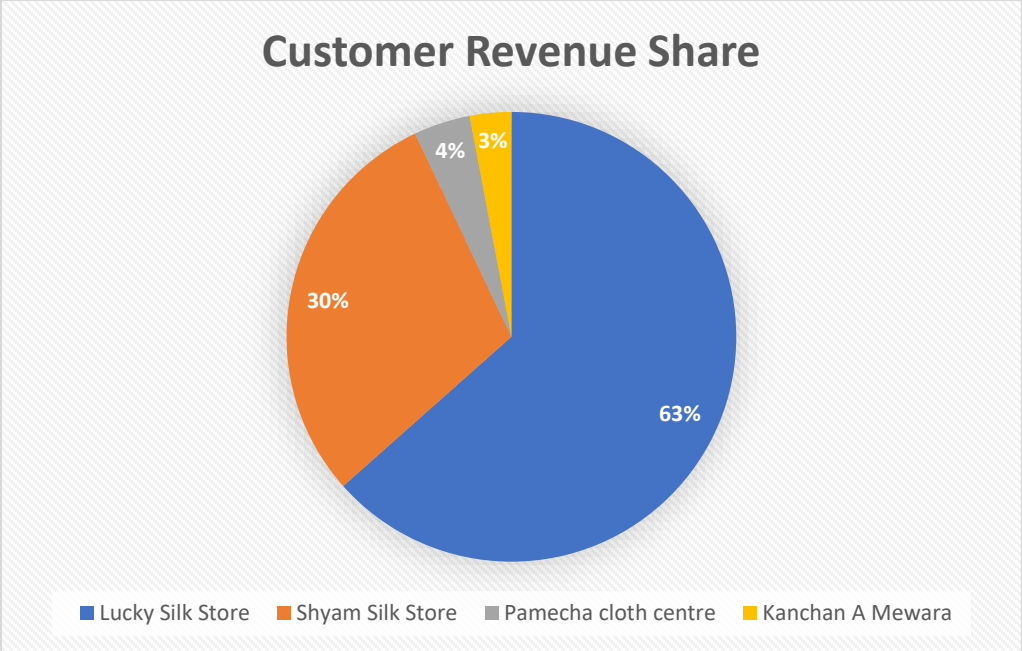


Figure 4: Customer Revenue share (Jan–Mar 2025)

Lucky Silk Store dominates at 63.4% of total revenue (₹3,64,455), followed by Shyam Silk Store at 29.5% (₹1,69,515). Pamecha Cloth Centre and Kanchan A Mewara contribute marginally. This extreme revenue concentration in a single customer poses a significant business risk: any payment delays or order reduction from Lucky Silk Store could severely impact the company's cash flow and profitability.

Expanding the active customer base and reducing dependence on a single buyer is a key recommendation for the final report.

5.5 Purchase vs. Sales Quantity by Category

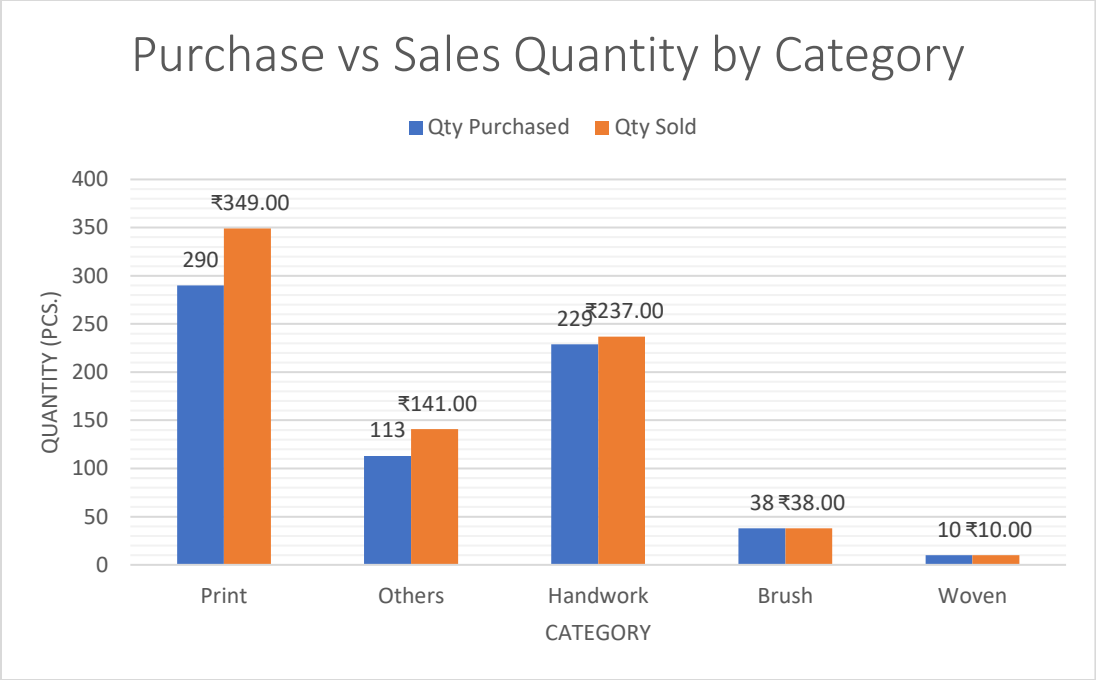


Figure 5: Purchase vs. Sales Quantity by Category (Jan–Mar 2025)

Comparing purchase and sales quantities across categories reveals interesting patterns. The Print category shows the largest absolute gap: 290 purchased vs. 349 sold, suggesting that previously held stock was drawn upon to fulfill sales — a positive inventory turnover signal. Handwork (229 purchased vs. 237 sold) and Others (113 purchased vs. 141 sold) also show sales exceeding purchases, indicating strong sell-through rates. Brush and Woven categories are balanced or slightly in surplus.

Overall, the business demonstrated a healthy inventory turnover during this period, with no category showing a severe buildup of unsold stock. However, the reliance on pre-existing inventory for Print items suggests restocking planning needs to be proactive.

5.6 **Conclusion**:

This mid-term report has presented a comprehensive data-driven analysis of S.R. Trading Co.'s sales and purchase operations over January to March 2025. Through systematic collection and analysis of transactional data from physical bills, the study provides meaningful insights into the business's revenue drivers, inventory dynamics, and customer structure.

The key findings from this mid-term analysis are as follows:

• The Print category is the dominant revenue driver, contributing 36.5% of total sales, and should remain the primary focus for restocking and supplier negotiations.

• February 2025 was the strongest sales month, suggesting a seasonal demand peak that should be anticipated in future inventory planning.

• Saree (EBR) is the most valuable individual SKU by revenue, and ensuring its consistent availability is critical to maintaining revenue performance.

• Over 63% of revenue is concentrated in a single customer — Lucky Silk Store — representing a significant credit and demand risk that the business must address through customer diversification.

• Purchase volumes were well-aligned with sales in most categories, indicating reasonably efficient inventory management for this period, though proactive restocking of high-velocity products remains an improvement area.

With the insights gathered in this mid-term phase, the final report will provide detailed and actionable recommendations on inventory replenishment models, credit policy improvements, customer acquisition strategies, and product mix optimization. These recommendations are aimed at improving S.R. Trading Co.'s profitability, operational efficiency, and long-term sustainability in the highly competitive Chandni Chowk saree market.